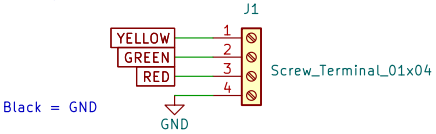
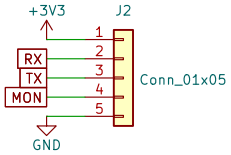
ECP (Honeywell/Ademco) interface – BJT version

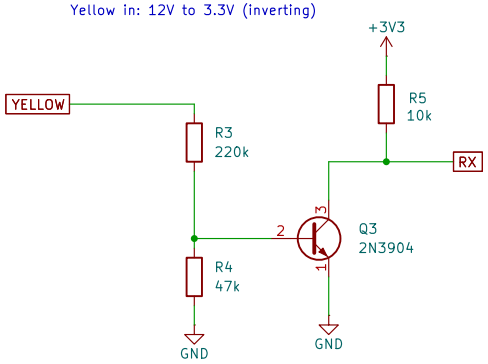
J1: 4–position 5.08 mm screw terminal, 22–16 AWG alarm wire



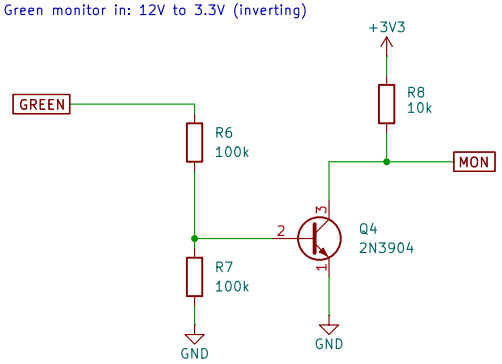




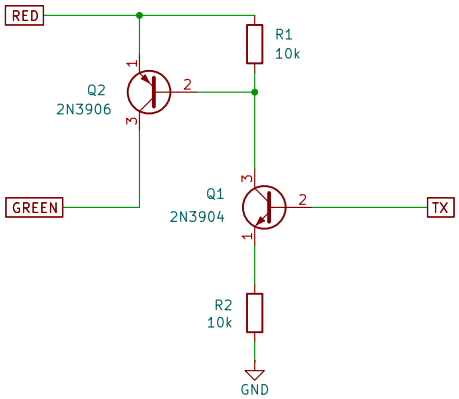
Yellow in: 12V to 3.3V (inverting)

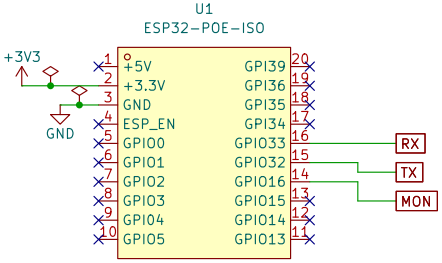


Green monitor in: 12V to 3.3V (inverting)



Green out: 3.3V to 12V





YAML config:

```
  invert_mon: "false"
  invert_rx: "false"
  invert_tx: "true"
  input_mode_rx: INPUT
  input_mode_mon: INPUT

  #ESP32–POE–ISO
  rxpin: 33
  txpin: 32
  monitorpin: 16
```

Since the internal logic already inverts the incoming data the above settings will be the reverse to what the circuit does.

Notes:

i: resistors r5,r8 and the 3.3v connection can be omitted but you will need to set:
   input_mode_rx: INPUT_PULLUP
   input_mode_mon: INPUT_PULLUP

ii: q1,q2,q3 and q4 can be substituted for other common small signal transistors.

iii: you can substitute resistor values but keep it within about 10% of the value up or down. You may go down to 4.7k for all 10k resistors.

iv: Q2 2N3906 PNP, Q1/Q3/Q4 2N3904 NPN. TO–92 pin order E–B–C assumed, check datasheet.

v: RX and MON are pulled up to the ESP32 3.3V by R5 and R8.
   Yellow idles high (panel 10k pull–up, low ~0.7V).
   Green idles low (panel 10k pull–down, 0V).

vi: j3 provides 12v output from the panel that can be used by an external step down to USB converter for powering the ESP.
  Powering the esp directly on the 5v pin is not recommended as it will conflict with the usb input if a cable is plugged in.